

Quantum Computing and Quantum Machine Learning

QNN, VQE, QAOA, QSVM, HHL

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Outline

Week 14 — Plan

- 1 Discussion of the **QAOA algorithm** with repetition from last week
- 2 Parametrized quantum circuits (PQC) and Variational Quantum Circuits (VQCs)
- 3 **Quantum Neural Networks (QNNs)**: mathematical structure, QFI, TDVP, barren plateaus, quantikz circuit examples
- 4 **Variational Quantum Eigensolver (VQE)**: variational ansatz, UCC, connections to many-body theory
- 5 **QAOA**: adiabatic motivation, MaxCut, parameter landscape
- 6 **QSVM**: quantum kernels, fidelity, RKHS
- 7 **HHL algorithm**: solving linear systems, QPE, Green's functions
- 8 Grand unification: Geometry + Dynamics + Operators

What is Quantum Machine Learning?

Quantum Machine Learning (QML) integrates quantum computing with machine learning to exploit quantum advantages:

- **Quantum parallelism:** process exponentially many states
- **Quantum entanglement:** correlated, non-classical feature spaces
- **Quantum interference:** enhance solutions, suppress errors

NISQ challenges

- Noisy Intermediate-Scale Quantum devices
- Decoherence limits circuit depth
- Efficient classical data encoding is non-trivial
- Scalability to large datasets is open

Core question

When and how does quantum hardware give a genuine advantage over the best classical algorithm?

QNN as a Variational Quantum State

The most compact mathematical description of a QNN:

$$|\psi(x, \theta)\rangle = U(\theta) U(x) |0\rangle^{\otimes n}$$

$$f(x, \theta) = \langle \psi(x, \theta) | O | \psi(x, \theta) \rangle$$

Three components

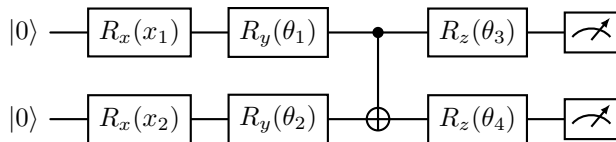
- $U(x)$: **feature map** — encodes classical data
- $U(\theta)$: **variational ansatz** — trainable unitary
- O : **observable** — defines the scalar output

Key point

Training a QNN means finding θ^* that minimises a classical loss $\mathcal{L}(\theta)$ computed from repeated quantum measurements.

Generic 2-Qubit QNN Circuit

$$|\psi(x, \theta)\rangle = W(\theta) U(x) |00\rangle$$



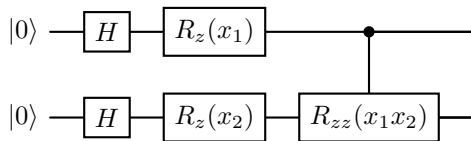
- **Layer 1** ($U(x)$): $R_x(x_i)$ encodes data as rotation angles
- **Layer 2** ($W_1(\theta)$): $R_y(\theta_i)$ variational rotations $\rightarrow$ CNOT entangler
- **Layer 3** ($W_2(\theta)$): $R_z(\theta_i)$ refines the state
- **Output:** $f = \langle Z_0 \rangle$ (expectation of $Z \otimes I$)

Feature Map as Hamiltonian Evolution

A common, physically motivated feature map:

$$U(x) = e^{-iH(x)}, \quad H(x) = \sum_i x_i Z_i + \sum_{i < j} x_i x_j Z_i Z_j$$

- Data $\equiv$ coupling constants of a spin Hamiltonian
- Circuit $\equiv$ quantum time evolution
- Pairwise terms create entanglement between features



$R_{zz}(\phi) = e^{-i\phi Z \otimes Z/2}$: two-qubit entangling gate encoding pairwise product $x_1 x_2$.

Pauli Expansion of the Variational Ansatz

The variational unitary is generated by a Pauli Hamiltonian:

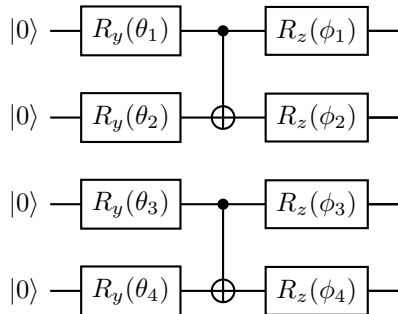
$$U(\theta) = e^{-iH(\theta)}, \quad H(\theta) = \sum_{\alpha} \theta_{\alpha} P_{\alpha}$$

where $P_{\alpha} \in \{I, X, Y, Z\}^{\otimes n}$ are Pauli strings.

The BCH expansion shows what the circuit computes:

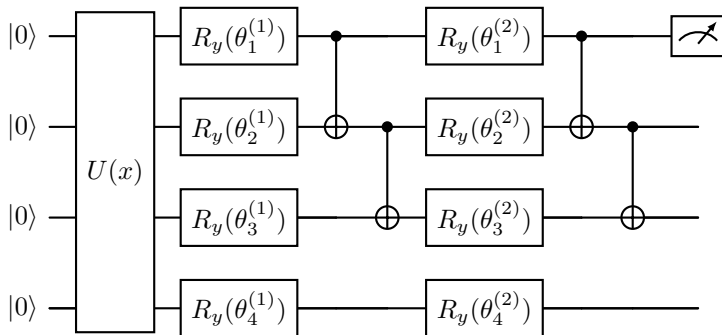
$$U^{\dagger} O U = O + i[H, O] + \frac{i^2}{2}[H, [H, O]] + \dots$$

Each commutator level adds higher-body correlations — analogous to coupled-cluster expansions.



One layer of a hardware-efficient ansatz (4 qubits): R_y rotations + CNOT ladder + R_z rotations.

Multi-Layer QNN: Full Circuit Diagram



Feature map $U(x)$ encodes data ($n = 4$ qubits); $L = 2$ repeated variational layers $W^{(l)}(\theta^{(l)})$ build up correlations via R_y rotations and CNOT ladders; qubit 0 is measured to give the scalar output $f = \langle Z_0 \rangle$.

Parameter-Shift Rule: Exact Quantum Gradients

For a gate $e^{-i\theta P/2}$ generated by a Pauli P (eigenvalues $\pm\frac{1}{2}$), the gradient is *exact*:

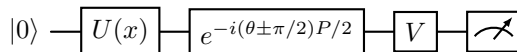
$$\frac{\partial f}{\partial \theta} = \frac{1}{2} [f(\theta + \frac{\pi}{2}) - f(\theta - \frac{\pi}{2})]$$

- Requires only 2 circuit evaluations per parameter
- Exact (not a finite-difference approximation)
- Hardware-friendly: no ancillas needed

Gradient derivation from commutator:

$$\partial_{\theta} f = \langle \psi | i[H_{\theta}, O] | \psi \rangle$$

Physical meaning: gradients are quantum response functions.



Two-point formula

Evaluate the circuit at $\theta + \frac{\pi}{2}$ and $\theta - \frac{\pi}{2}$; their half-difference gives the exact partial derivative.

Quantum Fisher Information Matrix

The **Quantum Fisher Information** (QFI) defines the Riemannian metric on the manifold of variational states:

$$F_{ij} = 4 \operatorname{Re}(\langle \partial_i \psi | \partial_j \psi \rangle - \langle \partial_i \psi | \psi \rangle \langle \psi | \partial_j \psi \rangle)$$

Information-geometric metric

$$ds^2 = \sum_{ij} F_{ij} d\theta_i d\theta_j$$

Measures the *distinguishability* of nearby quantum states — the Fubini-Study metric on projective Hilbert space.

Practical formula (overlap method)

$$F_{ij} = 4 \operatorname{Re} [\langle \partial_i \psi | (I - |\psi\rangle \langle \psi|) | \partial_j \psi \rangle]$$

Natural gradient descent

The *natural gradient* corrects the Euclidean gradient for the curved geometry of Hilbert space:

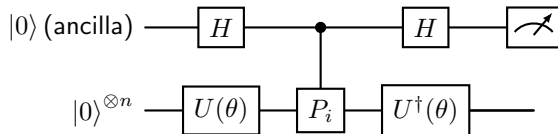
$$\dot{\theta} = F^{-1} \nabla_{\theta} \mathcal{L}$$

This is the basis of **Stochastic Reconfiguration (SR)** in VMC and of imaginary-time TDVP.

Diagonal elements: F_{ii} = variance of the Pauli generator P_i in state $|\psi\rangle$.

Computing the QFI: Hadamard-Test Circuit

The overlap $\langle \partial_i \psi | \partial_j \psi \rangle$ entering F_{ij} is measured by a Hadamard-test circuit:



Outcome probability:

$$P(0) = \frac{1}{2} + \frac{1}{2} \operatorname{Re} \langle \psi | P_i | \psi \rangle \implies F_{ii} = 4 \left(\langle \psi | P_i^2 | \psi \rangle - \langle \psi | P_i | \psi \rangle^2 \right) = 4 \operatorname{Var}(P_i)$$

- For off-diagonal elements F_{ij} use two Hadamard tests with $P_i \pm P_j$
- The QFI upper-bounds the classical Fisher information by the quantum Cramér-Rao inequality:
 $\operatorname{Var}(\hat{\theta}) \geq 1/F$

Time-Dependent Variational Principle (TDVP)

The TDVP projects the Schrödinger equation onto the tangent space of the variational manifold:

$$\delta \|(i\partial_t - H)|\psi\rangle\| = 0$$

Projecting $\langle \partial_i \psi | (i\partial_t - H) | \psi \rangle = 0$:

$$\sum_j F_{ij} \dot{\theta}_j = C_i, \quad C_i = \text{Im} \langle \partial_i \psi | H | \psi \rangle$$

Interpretation

QNN training $\equiv$ classical dynamical system on a curved manifold. Parameters θ_j evolve like generalised coordinates; F_{ij} is the mass matrix.

Imaginary-time TDVP

Replace $t \rightarrow -i\tau$ (imaginary time):

$$\sum_j F_{ij} \dot{\theta}_j = E_i$$

This is **Stochastic Reconfiguration** — optimal ground-state search on the variational manifold.

Barren Plateaus

The gradient variance scales exponentially:

$$\text{Var}(\nabla_{\theta} \mathcal{L}) \sim \frac{1}{2^n}$$

Why? Random, deep circuits form approximate **unitary 2-designs** — their output distribution concentrates near the Haar measure. Local observables concentrate around their mean, making gradients vanish.

Strategies to mitigate:

- Local cost functions (support on $O(1)$ qubits)
- Problem-inspired ansätze (exploit structure)
- Layer-wise training (incremental depth)
- Limiting entanglement in early training

Entanglement regimes

Entanglement	Training
Low	Classically simulable
Moderate	Optimal learning
High	Barren plateau

Quantum Neural Tangent Kernel

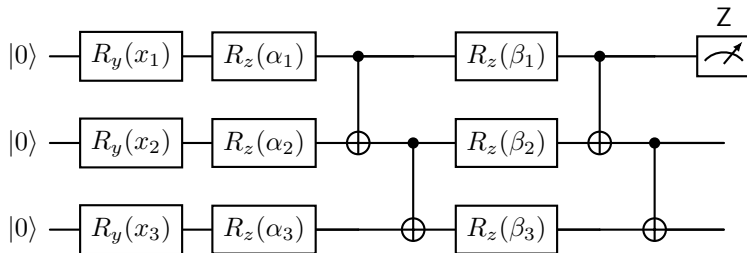
In the linearised training regime:

$$K(x, x') = \sum_i \frac{\partial f(x)}{\partial \theta_i} \frac{\partial f(x')}{\partial \theta_i}$$

connects QNNs to kernel methods.

Variational Quantum Classifier: Full Circuit

Three-stage circuit for binary classification on $n = 3$ qubits:



- **Encoding** (col. 2): $R_y(x_i)$ angle-embeds each feature
- **Var. layer 1** (cols. 3–5): R_z rotations + CNOT chain $0 \rightarrow 1 \rightarrow 2$
- **Var. layer 2** (cols. 6–8): second block of R_z + CNOT chain
- **Output:** $f = \langle Z_0 \rangle$; class 1 if $f > 0$, class 0 if $f < 0$

Entanglement Entropy and Expressivity

Entanglement entropy of subsystem A :

$$S(\rho_A) = -\text{Tr}(\rho_A \log \rho_A)$$

Controls the complexity of the variational state:

- $S = 0$: product state, no correlations, classically efficient
- $S = O(\text{area law})$: MPS-representable
- $S = O(n)$: volume-law, generic circuit output

The QNN ansatz lives on a *low-dimensional submanifold* of the full exponentially-large Hilbert space. The manifold dimension is $\dim \theta$.

Expressivity bound

A circuit with L layers and n qubits can generate states with at most

$$S(\rho_A) \lesssim L \log 2$$

entanglement entropy between any bipartition — a property used to prove simulability of shallow circuits.

Optimal regime

Moderate S (moderate entanglement): rich correlations without concentration of measure.

Simulating a QNN: NumPy + PyTorch

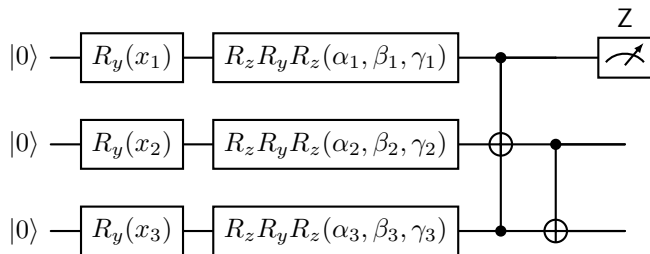
We simulate the circuit as $2^n \times 2^n$ unitary matrix operations in NumPy; gradients use the parameter-shift rule; optimisation uses PyTorch Adam.

```
def var_layer(p):
    """Ry(p0) ⊗ Ry(p1) → CNOT → Rz(p2) ⊗ Rz(p3)."""
    return kron(rz(p[2]), rz(p[3])) @ CNOT @ kron(ry(p[0]), ry(p[1]))

def circuit(params, x):
    """Full 2-qubit VQC: Rx feature map + 2 variational layers."""
    state = |00> # complex vector, length 4
    U = kron(Rx(x[0]), Rx(x[1])) # feature map
    U = var_layer(params[:4]) @ U # layer 1
    U = var_layer(params[4:8]) @ U # layer 2
    psi = U @ state
    return <psi| Z0 |psi> # expectation value

def grad_circuit(params, x, shift=pi/2):
    """Exact gradient via parameter-shift rule."""
    g = zeros(8)
    for i in range(8):
        g[i] = 0.5*(circuit(params + shift*e_i, x)
                    - circuit(params - shift*e_i, x))
    return g
```

Credit Classifier: 3-Qubit Strongly-Entangling Circuit



Encoding: $R_Y(x_i)$ angle-embeds income, debt ratio, age (scaled to $[0, \pi]$). Each feature becomes a rotation angle on one qubit.

Ansatz: general $SU(2)$ Euler rotation per qubit ($R_z R_y R_z = 3$ parameters per qubit), then *ring* of CNOTs ($0 \rightarrow 1 \rightarrow 2 \rightarrow 0$). Total: 9 trainable parameters.

Variational Quantum Eigensolver (VQE)

Variational principle:

$$E_0 = \min_{\psi} \langle \psi | H | \psi \rangle, \quad E(\theta) \geq E_0$$

Ansatz: $|\psi(\theta)\rangle = U(\theta) |0\rangle$

Measurement: decompose $H = \sum_i c_i P_i$:

$$E(\theta) = \sum_i c_i \langle P_i \rangle$$

Each $\langle P_i \rangle$ measured on the quantum device.

Gradient via parameter-shift rule:

$$\frac{\partial E}{\partial \theta} = \frac{1}{2} \left[E\left(\theta + \frac{\pi}{2}\right) - E\left(\theta - \frac{\pi}{2}\right) \right]$$

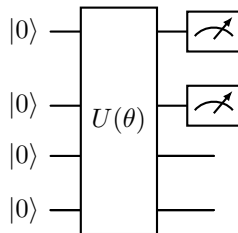
UCC ansatz:

$$U = e^{T-T^\dagger}, \quad T = \sum_{ai} t_{ai} a_a^\dagger a_i + \dots$$

Trotterised as $e^{A+B} \approx e^A e^B$.

BCH connection to coupled-cluster:

$$e^{-T} H e^T = H + [H, T] + \frac{1}{2} [[H, T], T] + \dots$$



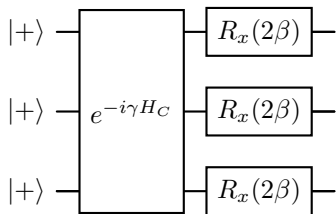
QAOA: Problem Statement and Ansatz

Goal: maximise a classical cost $C(z)$ over bit strings $z \in \{0, 1\}^n$.

Encode as a diagonal Hamiltonian: $H_C |z\rangle = C(z) |z\rangle$.

QAOA ansatz (depth p):

$$|\gamma, \beta\rangle = \prod_{l=1}^p e^{-i\beta_l H_M} e^{-i\gamma_l H_C} |+\rangle^{\otimes n}$$



One QAOA layer ($p = 1$): cost unitary then mixer.

- $H_M = \sum_i X_i$: mixer Hamiltonian
- $|+\rangle = H |0\rangle$: uniform superposition
- Maximise $C(\gamma, \beta) = \langle H_C \rangle$ over $2p$ parameters

Adiabatic connection

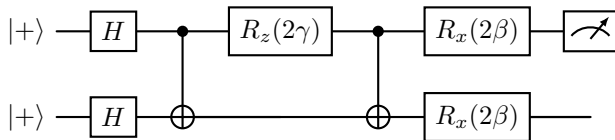
QAOA = Trotterised adiabatic evolution. $p \rightarrow \infty$ recovers exact adiabatic QC.

QAOA: MaxCut and Trotterisation

MaxCut on graph $G = (V, E)$: cost Hamiltonian

$$H_C = \sum_{\langle ij \rangle \in E} \frac{1 - Z_i Z_j}{2}$$

The cost unitary $e^{-i\gamma H_C}$ decomposes into two-qubit R_{zz} gates:



$p = 1$ QAOA for a single edge (i, j) : CNOT- R_z -CNOT implements $e^{-i\gamma Z_i Z_j / 2}$ on qubits i, j ; $R_x(2\beta)$ is the mixer.

Trotter expansion

$$e^{-i(H_M + H_C)t} \approx e^{-iH_M \Delta t} e^{-iH_C \Delta t}$$

QAOA at depth $p = p$ -step Trotterised adiabatic evolution with variable step sizes (γ_l, β_l) .

QAOA: Gradient, Landscape, and Relations to QNN/VQE

Gradient w.r.t. γ_l :

$$\partial_{\gamma_l} C = i \langle [H_C, H_{\text{eff}}] \rangle$$

Computed via *finite differences* (not parameter-shift, because H_C has a multi-valued spectrum $\{0, \dots, |E|\}$).

Landscape: $C(\gamma, \beta)$ non-convex but has structured correlations — warm-starting and parameter concentration exploit this.

Feature	QAOA	VQE
Ansatz	fixed	general
Motivation	adiabatic	variational
Target	opt. problem	ground state
Params	$2p$	many

QAOA $\subset$ QNN

QAOA is a QNN with *specific* Hamiltonians H_C, H_M instead of a general hardware-efficient ansatz.

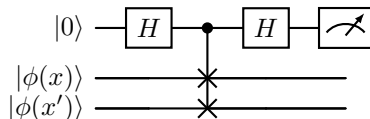
QSVM: Quantum Kernel Methods

Quantum feature map:

$$|\phi(x)\rangle = U(x) |0\rangle, \quad K(x, x') = |\langle\phi(x)|\phi(x')\rangle|^2$$

The kernel is a **fidelity** — a quantum overlap, directly analogous to many-body correlation functions.

Swap test circuit:



$$P(0) = \frac{1}{2}(1 + K(x, x'))$$

SVM dual optimisation

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2} \sum_{ij} \alpha_i \alpha_j y_i y_j K_{ij}$$

Solved classically once K_{ij} is estimated on the device.

Concentration risk

High entanglement in $U(x)$:

$$K(x, x') \approx \frac{1}{2^n} \quad \forall x, x'$$

Kernel becomes uninformative — avoid deep random feature maps.

HHL: Quantum Linear Systems Solver

Problem: given Hermitian A and $|b\rangle$, prepare $|x\rangle \propto A^{-1} |b\rangle$.

Spectral decomposition:

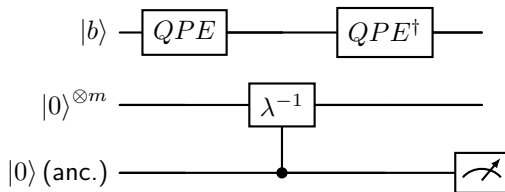
$$A^{-1} |b\rangle = \sum_j \frac{\beta_j}{\lambda_j} |u_j\rangle$$

HHL implements $\lambda_j \rightarrow \lambda_j^{-1}$ via QPE.

Steps:

- 1 QPE: encode eigenvalues in register
- 2 Controlled rotation: apply λ_j^{-1}
- 3 Uncompute QPE; post-select ancilla = $|1\rangle$

Complexity: $\mathcal{O}(\kappa^2 \log N)$ vs classical $\mathcal{O}(N\sqrt{\kappa})$.



Green's function connection

$$A^{-1} \sim (\omega I - H)^{-1} = G(\omega)$$

HHL $\equiv$ quantum computation of the resolvent operator / Green's function of the many-body Hamiltonian.

Three Fundamental Structures of QML

Quantum Machine Learning = Geometry + Dynamics + Operators

Structure	Mathematical object	QML method
Geometry	QFI metric F_{ij} , Hilbert space $\mathcal{H}$	QSVM (kernel = fidelity)
Dynamics	Variational evolution, TDVP	QNN, VQE, QAOA
Operators	Inverse maps, Green's functions	HHL, QPE

Unifying correspondences:

HHL $\longleftrightarrow$ exact inverse operator

QSVM $\longleftrightarrow$ kernel geometry $K(x, x') = |\langle \phi(x) | \phi(x') \rangle|^2$

QNN/VQE/QAOA $\longleftrightarrow$ variational dynamics $\sum_i F_{ij} \dot{\theta}_j = C_i$

Spectral Perspective and Learning as Projection

All methods rely on the spectral structure:

$$A = \sum_j \lambda_j |u_j\rangle \langle u_j|$$

- **HHL:** transforms $\lambda_j \rightarrow \lambda_j^{-1}$
- **QNN/VQE:** learn the eigenstructure via gradient descent on the variational manifold
- **QSVM:** measures pairwise overlaps between eigenstates

Learning as projection:

Learning $\equiv$ optimal projection onto
a submanifold of Hilbert space

Unified master equation

Learning $\equiv$ approximating operators

Approach	Operator
Exact	HHL (A^{-1})
Variational	QNN/VQE/QAOA
Geometric	QSVM

Information geometry:

$ds^2 = \sum_{ij} F_{ij} d\theta_i d\theta_j$; training follows geodesics of the QFI metric.

Outlook: Quantum Advantage and Future Directions

Where can genuine advantage emerge?

- **Fault-tolerant QC:** HHL, QPE, Shor — provable exponential speedup
- **NISQ VQA:** VQE, QAOA, QSVM — heuristic, unclear general advantage
- **Quantum kernels:** only when the feature map is classically hard to simulate

Fundamental trade-off

$\underbrace{\text{High expressivity}}_{\text{deep circuits}} \longleftrightarrow \underbrace{\text{Hard optimisation}}_{\text{barren plateaus}}$

Future directions:

- Fault-tolerant QC and error correction
- Adaptive ansatz design (ADAPT-VQE)
- Quantum-classical co-design (hardware-aware)
- Connection to many-body physics: VQE $\leftrightarrow$ coupled cluster, QAOA $\leftrightarrow$ adiabatic, HHL $\leftrightarrow$ Green's functions
- Quantum transformers and quantum RL
- Field-theoretic view: circuits as discretised quantum field theories

QNN and VQC

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